

# From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

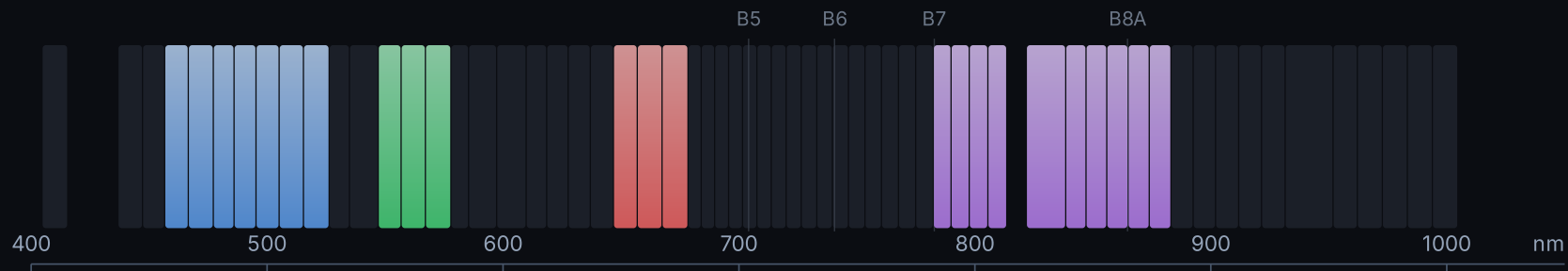
**which** each CHRIS band goes to the Sentinel-2 band it overlaps the most; in the NIR the model also averaged the red-edge band beside it

**how** plain **mean** of the CHRIS bands assigned to each pseudo-band

**then** DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

Training mapping, the one the model actually learned. The HuggingFace loader will be aligned to this.

# Mode 1 : 62 bands, ~34 m



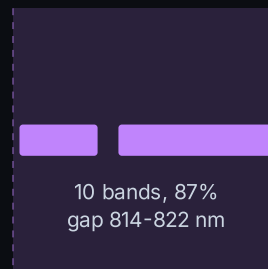
**B2**  
459-525 nm



**B3**  
542-578 nm

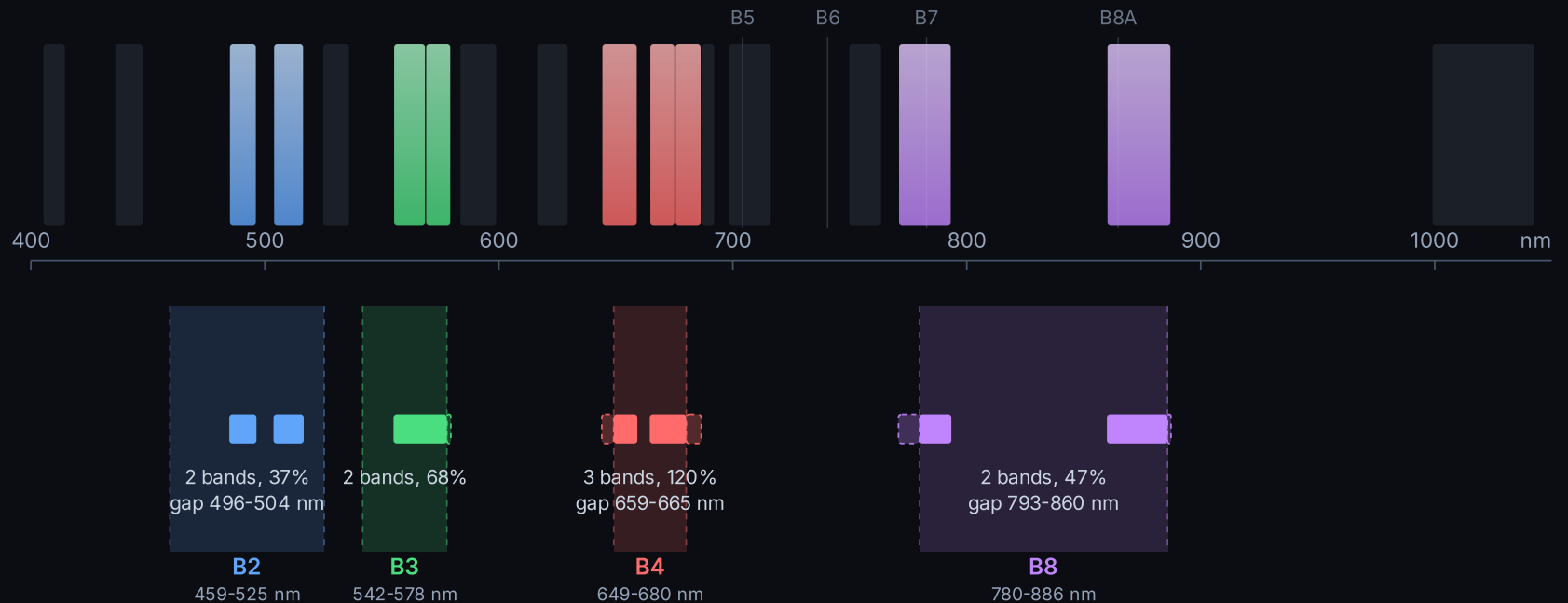


**B4**  
649-680 nm

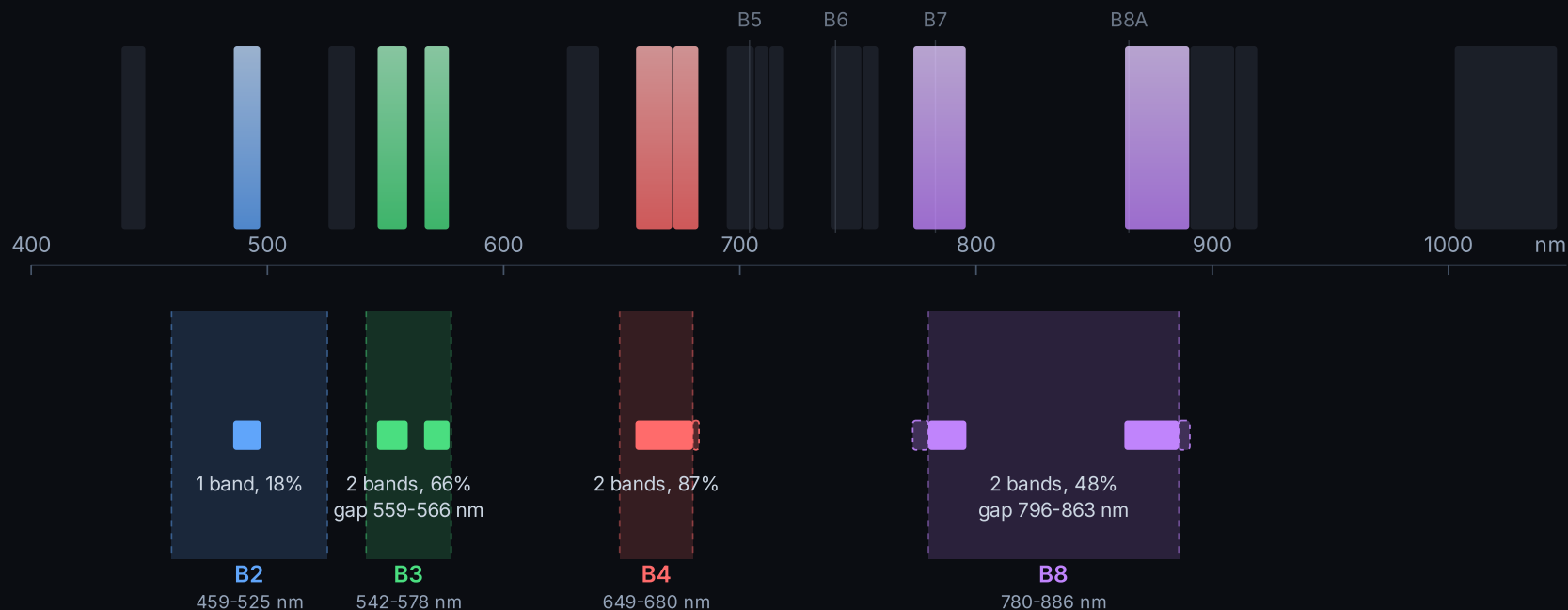


**B8**  
780-886 nm

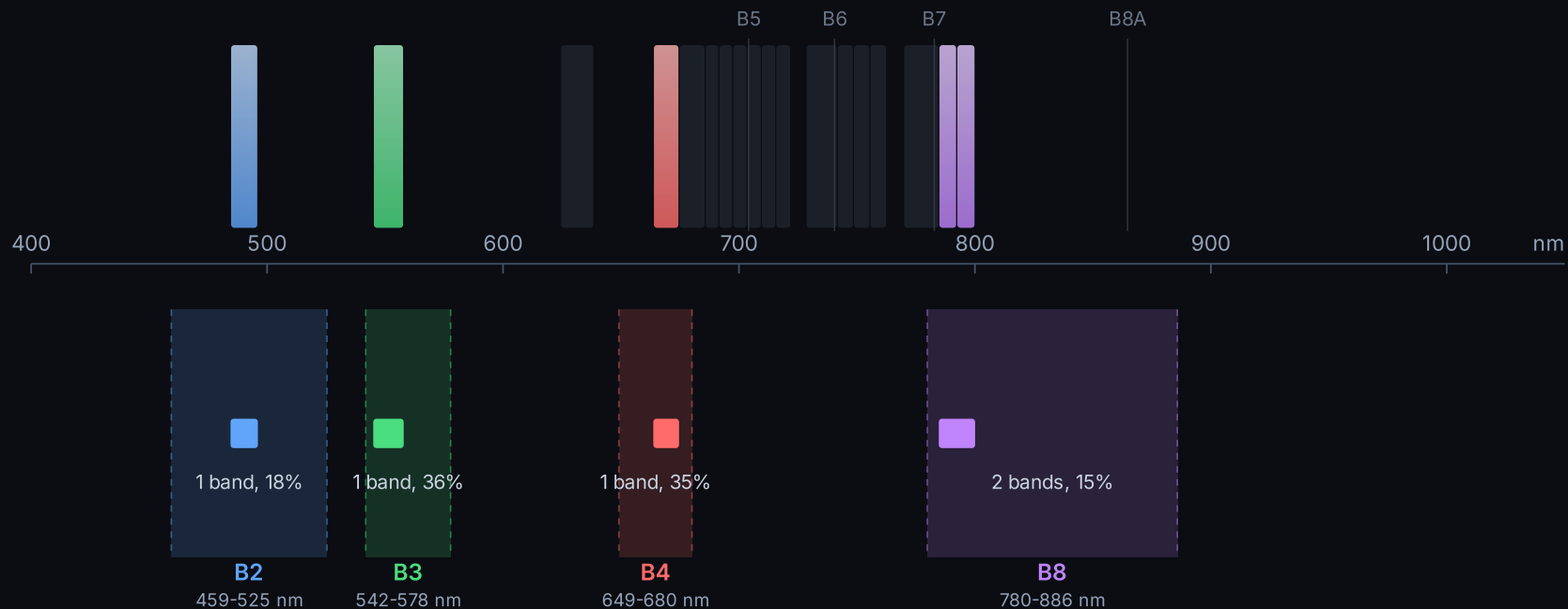
# Mode 2 : 18 bands, ~17 m



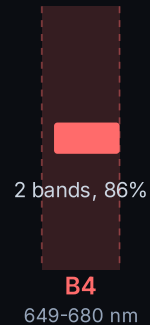
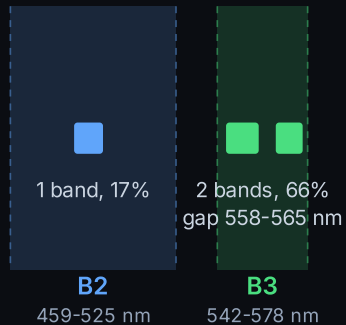
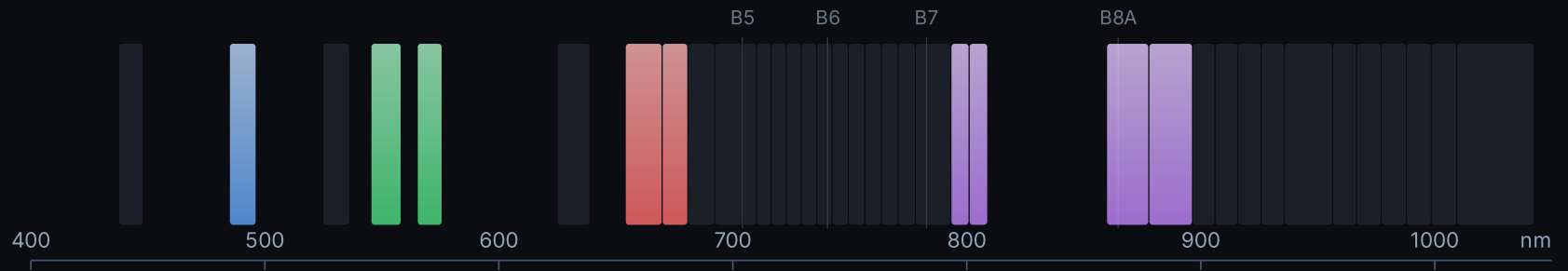
# Mode 3 : 18 bands, ~13 m



# Mode 4 : 18 bands, ~11 m



# Mode 5 : 37 bands, ~20 m



# Mode 20 : 4 bands, ~17 m, DN only



# Where the bands end up

R, B4

G, B3

B, B2

NIR, B8

| Mode | B2 blue              | B3 green           | B4 red             | B8 NIR              |
|------|----------------------|--------------------|--------------------|---------------------|
| 1    | 4-10 (7), 457-526 nm | 13-15 (3), 547-578 | 23-25 (3), 647-678 | 42-51 (10), 782-883 |
| 2    | 3-4 (2), 485-517     | 6-7 (2), 555-580   | 10-12 (3), 644-686 | 16-17 (2), 771-887  |
| 3    | 2 (1), 486-497       | 4-5 (2), 546-577   | 7-8 (2), 656-683   | 14-15 (2), 773-890  |
| 4    | 1 (1), 484-496       | 2 (1), 545-558     | 4 (1), 664-675     | 17-18 (2), 785-800  |
| 5    | 2 (1), 485-496       | 4-5 (2), 545-576   | 7-8 (2), 654-681   | 23-26 (4), 793-896  |
| 20   | 1                    | 2                  | 3                  | 4                   |

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.